



Miguel Pedro Peidro Paredes

AI SECURITY ARCHITECT · SECURE GENAI & AGENTIC SYSTEMS

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🌐 in/miguelpeidroparedes 🌐 Portfolio



FULL PORTFOLIO

Security architect for BBVA's GenAI programme since the first AI project reached the bank. I pair adversarial-ML research (GAN attacks on biometrics, first author at ACM ARES) with enterprise delivery: threat models, audits and the automations that keep regulated teams fast. I build agents myself, so I know where they break.

ARES '23

ACM · FIRST AUTHOR
GAN-synthesised MasterPrints vs. biometric security

~100k

PERSON-HOURS / YEAR
Saved by the automations I built at BBVA CIB

245+

AGENT CAPABILITIES
Tested on my own hybrid cloud/local agentic platform

\$16M

MARKET CAP AT 21
Co-founded the first DeFi gaming ecosystem (2020)

EXPERIENCE

AI Security Architect & IT Security Auditor

Feb 2024 – Present

BBVA · Corporate & Investment Banking · via Nfoque Advisory Services

- Security architect for BBVA CIB's GenAI and agentic initiatives since the start of AI adoption in the bank: security-by-design, threat modelling and risk sign-off for cloud-native AI architectures on AWS and Azure.
- Lead security audits and mitigation plans for critical banking systems, aligned with DORA, NIST and ISO 27001.
- Designed and shipped internal automations that save an estimated ~100,000 person-hours per year across CIB teams.

AI Researcher & Engineer · Guest Lecturer

Jul 2022 – Feb 2024

Universidad Carlos III de Madrid

- Built GAN pipelines that synthesise fingerprint patterns (MasterPrints) and tested them against real Android capacitive sensors; first author, ACM ARES 2023.
- Designed and taught AI-security training for INCIBE (Spanish National Cybersecurity Institute) and the Spanish National Police.

Cybersecurity & Distributed Systems Analyst

Dec 2021 – Mar 2023

FUNDITEC

- Engineered secure distributed architectures and smart contracts for Horizon 2020 and NextGenerationEU projects (supply-chain and renewable-energy traceability, Gaia-X, Peppol).

Co-founder & IT Security Manager

Dec 2020 – Aug 2021

DFSocial Gaming · DecodedWorld

- Co-founded at 21 and secured the first DeFi gaming ecosystem on Ethereum and BNB Smart Chain; peaked at a \$16M market cap with a 16-person team.

EDUCATION

Doctoral research, AI × Cybersecurity · UC3M

2022 – 2024 · paused

Thesis topic: the MasterPrint threat on Android capacitive sensors

MSc Cybersecurity · UC3M

2020 – 2021

Thesis: analytical tool for Advanced Persistent Threats using geopolitical data (GDELT)

BSc Computer Engineering · UC3M

2016 – 2020

Thesis: software for malicious macro detection

LANGUAGES

Spanish (native) · English (C1, professional) · French (A2)

SELECTED PROJECT

Hybrid Agentic Platform

Independent project

My own platform for running agentic systems in corporate environments across cloud and local infrastructure, designed to minimise the cost of adopting agents. Its agents cover 245+ tested technical and operational capabilities.

RESEARCH

“Characterizing the MasterPrint threat on Android devices with capacitive sensors”

ACM ARES 2023 · First author

doi.org/10.1145/3600160.3600177

CONTINUOUS LEARNING

CMU 11-768 · AI Agents

ONGOING

Berkeley · Agentic AI

AI Security Bootcamp (AISB)

BlueDot · Technical AI Safety

Stanford CS224W · ML with Graphs

Stanford · Cryptography I

UIUC · Cloud Computing Concepts 1-2

Helsinki · Big Data Platforms

SKILLS

SECURITY

Security-by-design for GenAI & agents · Threat modelling & risk · Security audits · Adversarial ML · DORA · NIST · ISO 27001 · GDPR

AI & AGENTS

LLM agents & multi-agent systems · Hybrid cloud/local inference · RAG & tool use · GANs & neural networks · Process automation

PLATFORMS

AWS · Azure · Ethereum / BNB Chain · Smart contracts